Programme Name/s	**: Automation and Robotics/ Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Medical Electronics/ Electronics & Computer Engg.**
Programme Code	**: AO/ DE/ EJ/ EK/ ET/ EX/ IC/ IE/ IS/ MU/ TE**
Semester	**: Second**
Course Title	**: BASIC ELECTRONICS**
Course Code	**: 312314**

I. RATIONALE

Diploma engineers must deal with the various electronic components while maintaining various electronic equipment/systems. The use of basic electronics components and handling of various electronics systems will help them troubleshoot electronics equipment used in industry or in the consumer market etc. This course is developed to empower the students to apply their knowledge to solve broad electronic engineering application problems.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to attend following industry identified competency through various teaching learning experiences: • Maintain electronic equipment/systems comprising of discrete electronic components.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use relevant diode in electronics circuits.
- CO2 - Use BJT in electronics circuits .
- CO3 - Use of BJT as amplifier and switch ..
- CO4 - Use FET and MOSFET in electronics circuits..
- CO5 - Maintain DC regulated power supply.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
312314	BASIC ELECTRONICS	BEL	AEC	4	-	4	-	8	4	3	30	70	100	40	50	20	25@	10	-	-	175

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe working principle, characteristics, and application of the given type of diode. TLO 1.2 Describe the working of given type of rectifier. TLO 1.3 Calculate ripple factor, PIV, and efficiency of the given type of filter. TLO 1.4 Describe the need and working of rectifier filter circuit.	**Unit - I Applications of Diode** 1.1 Different types of diodes and their materials: Construction, Symbol, working principle, applications, Forward and reverse biasing and V-I characteristics of following diodes: P-N junction diode, Zener diode, LED, Photo diode, Schottky diode, 1.2 Diode as rectifier: Types of Rectifiers, Half wave, Full wave (bridge rectifier and center tapped), circuit operation, Input- output waveform for voltage and current, Parameters of rectifier: Average DC value, value of current and voltage, ripple factor, ripple frequency, PIV of diode, TUF, efficiency of rectifier. 1.3 Types of Filters: Shunt capacitor, Series inductor, LC and CLC filter. 1.4 Rectifier IC – KBU 808 IC pin diagram and application .	Chalk-Board Video Demonstrations
2	TLO 2.1 Describe the working principle of the given type of transistor. TLO 2.2 Calculate current gain for given configuration of BJT TLO 2.3 Compare configuration of transistors. TLO 2.4 Justify the need of biasing method. TLO 2.5 Describe the procedure to minimize the thermal runaway effect.	**Unit - II Bipolar Junction Transistor** 2.1 Current operating device. 2.2 Different types of transistors: PNP, NPN. 2.3 Transistor configurations: CB, CE, CC Transistor characteristics (input, and output) in different transistor configuration. Relation between alpha ,beta, gama. Comparison between CB, CC and CE. 2.4 4 BJT biasing: Need of DC load Line, Operating point, stabilization, thermal runaway, heat sink. Types of biasing: fixed biasing, base bias with emitter feedback, voltage divider.	Chalk-Board Video Demonstrations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain with sketches the working principle of the given type of amplifier. TLO 3.2 Describe working of Single Stage Transistor Amplifier. TLO 3.3 Calculate Voltage gain and bandwidth TLO 3.4 Describe working of Multistage amplifiers TLO 3.5 Describe working of BJT as a Switch	**Unit - III BJT Amplifiers** 3.1 Classification of amplifier, BJT as an amplifier. 3.2 Single Stage Amplifier: Working , various currents (Ib, Ic,Ie), Voltage gain of CE amplifier (no derivations required), Frequency response of CE amplifier. Simple numericals. 3.3 Multistage amplifiers: General Multistage BJT based amplifiers 3.4 Types of BJT amplifier coupling: Circuit diagram, operation frequency response and applications of Direct coupled, RC coupled and transformer coupled. 3.5 BJT as a Switch	Chalk-Board Video Demonstrations
4	TLO 4.1 Explain the working of given type of FET TLO 4.2 Explain the given type of FET biasing method. TLO 4.3 Describe working of FET Amplifier. TLO 4.4 Explain working of given type of MOSFET. TLO 4.5 Differentiate working principle of FET and MOSFET on the basis of the given characteristics of curve.	**Unit - IV Field Effect Transistor** 4.1 Voltage operating device, Construction of JFET (N-channel and P- channel), symbol, working principle and characteristics (Drain and Transfer characteristics) , different parameters of FET . FET applications 4.2 FET Biasing: Source self-bias, drain to source bias. 4.3 Common source FET amplifier. 4.4 MOSFET: Construction, working principle and characteristics of Enhancement and depletion MOSFET, MOSFET handling.	Chalk-Board Video Demonstrations
5	TLO 5.1 Describe the working of the DC regulated power supply. TLO 5.2 Calculate output voltage of the given Zener voltage regulator circuit.. TLO 5.3 Describe the working of 78XX and 79XX fixed voltage IC Regulator. TLO 5.4 Describe the working of IC 723 as Low and High voltage regulator. TLO 5.5 Explain block diagram of Switch Mode Power supply.	**Unit - V Regulators and Power supply** 5.1 Need of Regulated power supply . Basic block diagram of DC regulated power supply and function of each block 5.2 Load and Line regulation. 5.3 Zener diode voltage regulator 5.4 Fixed voltage IC Regulator: Three terminal Pin diagram, working and application of 78XX and 79xx series. 5.5 Variable voltage IC Regulator : IC 723 pin diagram , block diagram, working. Low voltage regulator, High voltage regulator 5.6 Switch Mode Power supply : Need of SMPS , block diagram and functions of blocks.	Chalk-Board Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Test PN junction Diode in forward bias. LLO 1.2 Plot the V-I characteristics of PN junction diode and determine cut in voltage. LLO 1.3 Calculate static and Dynamic resistance of diode.	1	* Test the performance of PN Junction diode	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Test Zener Diode in reverse bias. LLO 2.2 Plot V-I characteristics of Zener Diode in reverse bias. .	2	* Test the performance of zener diode	2	CO1
LLO 3.1 Build the circuit for Photo Diode . LLO 3.2 Observe the change in current with change in light intensity of the source LLO 3.3 Plot distance VS Photo diode Current	3	* Check the performance of photo diode by varying the light intensity as well as the distance of the light source.	2	CO1
LLO 4.1 Construct the circuit for Half Wave Rectifier using PN junction Diode on. LLO 4.2 Plot Output Waveform for sinusoidal input.	4	* Construct and Test the half wave rectifier.	2	CO1
LLO 5.1 Build the circuit for Half Wave Rectifier with LC filter/ Pi filter using PN junction Diode. LLO 5.2 Obsrve and draw input & output waveforms for sinusoidal wave .	5	* Build and Test the half wave rectifier with LC filter/ π filter	2	CO1
LLO 6.1 Prepare the circuit for Full Wave Centre Tapped Rectifier using PN junction Diode. LLO 6.2 Observe and draw input & output waveform for sinusoidal wave.	6	* Prepare and Test the full wave rectifier using two diodes.	2	CO1
LLO 7.1 Build the circuit for Full Wave Bridge Rectifier using PN junction Diode LLO 7.2 Observe and draw input & output waveform for sinusoidal wave.	7	* Build and Test the full wave Bridge rectifier on bread board using two diodes.	2	CO1
LLO 8.1 Build the circuit for Full Wave Rectifier using PN junction Diode with LC/Pi filter. LLO 8.2 Calculate ripple factor for given setup.	8	* Use LC/ π filter with full wave rectifier to measure ripple factor	2	CO1
LLO 9.1 Construct the circuit for full wave rectifier using IC KBU 808 with filter LLO 9.2 Observe and draw input & output waveform for sinusoidal wave.	9	* Construct and Test the full wave rectifier on bread board using IC KBU 808 with filter.	2	CO1
LLO 10.1 Build the circuit for 7 Segment LED display FND 507/508. LLO 10.2 Observe numeric output for 0-9	10	Bulid and Test the performance parameters of 7 Segment LED display FND 507/508.	2	CO1
LLO 11.1 Identify the terminals of the PNP and NPN transistor for TO-5, TO-220, TO-66 LLO 11.2 Select of transistor for different max. voltage, current and switching speed	11	* Identify and select transistors using datasheets	2	CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 12.1 Build the circuit for BJT in common base configuration. LLO 12.2 Plot input and output characteristics of common base configuration.	12	Build and Test the performance of BJT working in CB mode.	2	CO2
LLO 13.1 Select the specific transistor for different max. voltage, current and switchingspeed LLO 13.2 Prepare the circuit for BJT in common emitter configuration.	13	* Prepare and Test the performance of BJT working in CE mode	2	CO2
LLO 14.1 Build the circuit for BJT voltage divider bias circuit. LLO 14.2 Locate Q point on Load line.	14	* Build and Test the BJT voltage divider bias circuit for given input	2	CO2
LLO 15.1 Test the performance parameters of BJT as Switch LLO 15.2 Identify Cutoff and saturation regions	15	* Construct and Test the performance parameters of BJT as Switch.	2	CO2
LLO 16.1 Build single stage Common emitter amplifier. LLO 16.2 Plot frequency response for Common emitter amplifier.	16	* Build and Test the performance of single stage Low Power Common emitter amplifier	2	CO3
LLO 17.1 Build the circuit for BJT common emitter (CE) amplifier using simulation software (like SPICE/Multisim) LLO 17.2 Plot Output Waveform for sinusoidal input. LLO 17.3 Plot frequncy response curve .	17	Simulate and Test output waveform and frequency response of single stage common emitter (CE) amplifier using simulation software (like SPICE / Multisim)	2	CO3
LLO 18.1 Build the circuit for BJT two stage RC coupled common emitter (CE) amplifier. LLO 18.2 Plot frequency response	18	* Build and Test the performance of RC coupled two stage amplifier.	2	CO3
LLO 19.1 Build the circuit for FET in common source configuration. LLO 19.2 Plot characteristics for drain to source voltage VDS verses drain current ID for different Values of VGS	19	* Test the performance of FET drain characteristics	2	CO4
LLO 20.1 Build the circuit for FET in common source configuration. LLO 20.2 Plot characteristics for Gate to source voltage VGS verses drain current ID LLO 20.3 Calculate transconductance.	20	* Check the performance of FET transfer characteristics and calculate transconductance	2	CO4
LLO 21.1 Build the circuit for FET in common source configuration. LLO 21.2 Plot characteristics for Gate to source voltage VGS verses drain current ID	21	* Build and Test the performance of common source FET amplifier	2	CO4
LLO 22.1 Test the voltages &waveforms at various Test points of regulated dc power supply.	22	Test the various blocks of regulated dc power supply.	2	CO5
LLO 23.1 Identify the various faults in the Regulated DC power supply.	23	* Find out faults at different stages of regulated dc power supply.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 24.1 Rectify the various faults in the Regulated DC power supply	24	* Trouble shoot given DC regulated power supply.	2	CO5
LLO 25.1 Construct Zener voltage regulator for given voltage. LLO 25.2 Calculate load and line regulation.	25	Construct and test the performance of Zener voltage regulator for given voltage.	2	CO5
LLO 26.1 Build the circuit for Positive voltage regulator using 78XX IC. LLO 26.2 Calculate load and line regulation.	26	* Build and Test the performance of Positive voltage regulator using 78XX , three terminal IC for given voltage.	2	CO5
LLO 27.1 Build the circuit for Negative voltage regulator using 78XX IC. LLO 27.2 Calculate load and line regulation.	27	Build and Test the performance of Negative voltage regulator using 79XX, three terminal IC for given voltage.	2	CO5
LLO 28.1 Construct the circuit for Dual voltage regulator using 78XX and 79XX IC. LLO 28.2 Calculate load and Line regulation.	28	* Construct and test the performance of Dual voltage regulator using 78XX and 79XX ,three terminal IC for given voltage	2	CO5
LLO 29.1 Build LOW voltage regulator circuit using IC LM723 (2V-7V). LLO 29.2 Calculate load and line regulation.	29	* Build and Test the performance of LOW voltage regulator using IC LM723 for given voltage.(2 V-7V)	2	CO5
LLO 30.1 Build High voltage regulator circuit using IC LM723 (7V-30V) LLO 30.2 Calculate load and line regulation.	30	Build and Test the performance of HIGH voltage regulator using IC LM723 for given voltage.(7V-30V)	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Build Audio amplifier using BJT.
- Build the circuit for 3v battery charger.
- Build Clap switch Using transistor.
- Build audio amplifier using IC LM386.
- Build power supply using LM317.
- Prepare a chart of different types of Rectifiers showing their specifications and applications

Assignment

- Study working of OLED display.
- study of different Audio amplifier ICs (min 4).
- Study working of MOSFET as variable capacitor.
- select specific FET and Study datasheet for same.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Variable DC Power supply 0-30V with display for voltage and current, 2Amp SC protection	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,19,20,22,23,24
2	LT Spice /Lab view/H Spice /P Spice /HS Spice / Multisim/ Proteus/Octeva or any other relevant open source software	17
3	Computer System with advanced Configuration Hardware requirement as per selected software	17
4	DSO 30/50/100 MHz Frequency Digital read out USB interface	4,5,6,7,8,9,16,22
5	CRO 20/30/100 MHz Frequency Dual Channel External Trigger CT mode facility or any other better specifications	4,5,6,7,8,9,16,22,18
6	Function Generator 0-2 MHz with Sine, square and triangular output with variable frequency and amplitude	4,5,6,7,8,9,16,22,18
7	Analog multimeter& Digital multimeter	All
8	Different types of cables and connectors	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Applications of Diode	CO1	12	4	4	6	14
2	II	Bipolar Junction Transistor	CO2	12	4	4	6	14
3	III	BJT Amplifiers	CO3	14	4	6	6	16
4	IV	Field Effect Transistor	CO4	12	4	6	4	14
5	V	Regulators and Power supply	CO5	10	4	4	4	12
Grand Total				**60**	**20**	**24**	**26**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Progrssive test ,Assignment, Microproject , Termwork
- Each practical will be assessed considering - - 60% weightage to process and - 40% weightage to product
- Continuous assessment based on process and product related performance indicators, laboratory experience.

Summative Assessment (Assessment of Learning)

- End of Term Examination, Laboratory performance.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	2	1	1	1	1	1			
CO2	2	2	1	1	1	1	1			
CO3	2	2	1	1	1	1	1			
CO4	2	2	1	1	1	1	1			
CO5	2	2	2	1	2	2	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Mehta, V.K. Mehta, Rohit Mehta	Principles of Electronics	S.Chand New Delhi, edition-2008 ISBN-13: 978- 8121927833
2	Sedha, R.S.	A Text book of Applied Electronics	S.Chand (G/L) & Company Ltd; ISBN-13 978-8121904209
3	P.Ramesh Babu	Electronics Device and Circuits	Scitech Publications (India) Pvt Ltd ,ISBN-13 978-8183712156
4	Theraja B.L. (Author), Sedha R.S. (Author)	Principles of Electronic Devices and Circuits (Analog and Digital)	S Chand & Company,ISBN-13 978-8121921992
5	B.L.Theraja	Basic Electronics (solid State)	S Chand;ISBN-13 978-8121925556
6	Albert P. Malvino, David J. Bates	Electronic Principles	McGraw Hill; ISBN-13 978-9354602399
7	D. P. Kothari , I. J. Nagrath	Basic Electronics	McGraw Hill Education,ISBN-13 978-9352606467
8	Roberrt L.Boylestead	Electronics Circuit and Circuit theory	Pearson Education India, ISBN-13 978-9332542600

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://nptel.ac.in/courses/122106025	Basic Electronics and Lab, IIT Madras Prof. T.S. Natarajan
2	https://archive.nptel.ac.in/courses/108/101/108101091/	Basic Electronics, IIT Bombay
3	4. https://learn.sparkfun.com/tutorials/transistors	Transistor basics
4	https://www.multisim.com/	online multisim live software/ free student evalution software download for limited time
5	https://alternativeto.net/software/multisim/	alernative softwares to multisim
6	https://www.labcenter.com/	demo version of Proteus software
7	https://learn.sparkfun.com/tutorials/transistors	Simulation

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BASIC ELECTRONICS **Course Code : 312314**

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

MSBTE Approval Dt. 01/10/2024 **Semester - 2, K Scheme**

Programme Name/s	**: Automation and Robotics/ Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Medical Electronics/ Electronics & Computer Engg.**
Programme Code	**: AO/ DE/ EJ/ EK/ ET/ EX/ IC/ IE/ IS/ MU/ TE**
Semester	**: Second**
Course Title	**: ELEMENTS OF ELECTRICAL ENGINEERING**
Course Code	**: 312315**

I. RATIONALE

A technical person has to deal with the various electrical machines, equipment, and protective devices. In order to increase the technical proficiency, a technician should possess essential knowledge of electrical engineering parameters, basic concepts, and laws of electrical engineering.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Use electrical equipment efficiently for different electronic engineering application.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Interpret the magnetic field parameters for the particular magnetic circuits.
- CO2 - Analyze A.C. circuits for single phase and polyphase supply.
- CO3 - Select the transformer and DC motor for the given application.
- CO4 - Select the fractional horse power motor for the given application.
- CO5 - Choose the protective devices for the electrical protection.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
312315	ELEMENTS OF ELECTRICAL ENGINEERING	EEE	SEC	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the terms related to Magnetic circuit. TLO 1.2 Distinguish between electric and magnetic circuit. TLO 1.3 Interpret magneto motive force in series and parallel magnetic circuit. TLO 1.4 Describe laws related to magnetic circuit. TLO 1.5 Classify the types of induced electromotive force	**Unit - I Magnetic circuits** 1.1 Define and state units of Magnetic flux, Flux density, Magnetomotive force, Magnetic field strength, Permeability. 1.2 Electric circuit and magnetic circuit analogy and differences. 1.3 Series and parallel magnetic circuit. 1.4 Faraday's laws of electro-magnetic induction, Lenz's law, Fleming right hand and left hand rule. 1.5 Dynamically and statically induced emf, self and mutual induced Electromotive force and its inductances.	Presentations Chalk-Board Video Demonstrations Model Demonstration

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Compare AC quantities with DC quantities. TLO 2.2 Describe terminology related to A.C. fundamentals. TLO 2.3 Describe different forms of representation for electrical quantity. TLO 2.4 Analyze A.C. circuits for different types of load. TLO 2.5 Explain generation of three phase induced emf. TLO 2.6 Analyze three phase circuit for star and delta connection.	**Unit - II A.C fundamentals for single phase and polyphase circuits** 2.1 Define A.C. and D.C. quantities, advantages of A.C over DC. 2.2 Single phase sinusoidal A.C. wave: instantaneous value, cycle, amplitude, time period, frequency, angular frequency, R.M.S. value, average value for sinusoidal waveform. 2.3 Vector, polar and complex forms representation of an ac quantity, phase angle, phase difference concept of lagging and leading. 2.4 A.C through pure resistance, inductance and capacitance. Its equation, vector diagram and waveform. 2.5 Define polyphase system and advantages of three phase system over single phase system. 2.6 Generation of three phase induced emf and its waveform. 2.7 Phase and line currents, phase and line voltages in star connected and delta connected balanced load system.	Video Demonstrations Presentations Chalk-Board
3	TLO 3.1 Explain construction and working principle of given type of transformer. TLO 3.2 Select different types of transformer for the particular application. TLO 3.3 Describe construction and the working of DC motor. TLO 3.4 Select the type of DC motor for given application.	**Unit - III Transformers and DC motors** 3.1 Transformer construction and working principle, emf equation, voltage ratio, transformation ratio. 3.2 Auto-transformer, Pulse transformer and Isolation transformer construction, working principle and applications. 3.3 DC motor construction and working principle. 3.4 Different types of DC motors with its schematic diagram. 3.5 Applications of DC motors.	Chalk-Board Model Demonstration Video Demonstrations Presentations
4	TLO 4.1 Explain the construction and working principle of the given type of FHP motor. TLO 4.2 Select relevant FHP motor for the respective application TLO 4.3 Describe the procedure to connect given motor for the given application.	**Unit - IV Fractional horse power motors** 4.1 Construction, working principle and application of split phase single phase AC induction motors. 4.2 Construction, working principle and application of universal motor and reversal of direction of rotation. 4.3 Construction, working principle and application of stepper motor. Only concept of speed control, stepper motor's reversal of direction of rotation 4.4 Construction, working principle, specification and application of linear induction motor	Model Demonstration Presentations Chalk-Board Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Explain general safety rule of electrical system. TLO 5.2 Explain and select the different types of protective devices. TLO 5.3 Draw circuit connection diagram of protective devices. TLO 5.4 Describe earthing system and related terms.	**Unit - V Electrical protective devices** 5.1 Electrical general safety rules, Personal Protective Equipment (PPE), Selection of wires and cable as per application. 5.2 Type of fuses, operation, connection diagram and application of fuses, Miniature Circuit Breaker(MCB), Moulded Case Circuit Breaker (MCCB), Earth Leakage Circuit Breaker(ELCB)operation, connection diagram and general specification 5.3 Draw circuit connection diagram of Protective devices. 5.4 Need of Earthing, methods of earthing, types of earthing and factors affecting earthing as per Indian Electricity rule.	Model Demonstration Video Demonstrations Presentations Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Use Faraday's law of electro-magnetic induction. LLO 1.2 Classify types of induced emf.	1	*Demonstration of Faraday's law of electro-magnetic induction for statically and dynamically induced emf.	2	CO1
LLO 2.1 Use Faraday's law of electro-magnetic induction. LLO 2.2 Observe mutual induced emf in transformer.	2	*Demonstration of Mutually induced EMF by using single-phase transformers.	2	CO1
LLO 3.1 Use cathode ray oscilloscope. LLO 3.2 Identify different parameters on CRO.	3	*Measure frequency, Time period, Peak value, RMS value of sinusoidal AC waveform using CRO.	2	CO2
LLO 4.1 Identify phase angle and phase difference of given quantities. LLO 4.2 Identify the nature of power factor for the respective circuit.	4	Observe the phase difference between voltage and current on CRO for resistive, inductive, and capacitive load and comment on the nature of the power factor (Lagging, Leading, Unity).	2	CO2
LLO 5.1 Connect star connected three phase load. LLO 5.2 verify relationship between line and phase quantities.	5	*Connect three phase star connected balanced load and verify the relationship between line voltage and phase voltage, line current and phase current.	2	CO2
LLO 6.1 Connect delta connected three phase load. LLO 6.2 verify relationship between line and phase quantities.	6	Connect three phase delta connected balanced load and verify the relationship between line voltage and phase voltage, line current and phase current.	2	CO2
LLO 7.1 Calculate transformation ratio of transformer. LLO 7.2 Connect transformer to given load.	7	*Determine the transformation ratio current ratio of single phase transformer.	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 8.1 Identify pin configuration of pulse transformer. LLO 8.2 Check electrical isolation between input and output of pulse transformer.	8	Demonstration of working of pulse transformer by observing input pulse and output pulse of pulse transformer on CRO.	2	CO3
LLO 9.1 Identify different parts DC motor. LLO 9.2 Identify different types of DC motor.	9	Identify different types of DC motor by observing terminal connections and also identify different parts of DC motor.	2	CO3
LLO 10.1 Connect DC motor to DC supply. LLO 10.2 Select particular starter for particular motor starting.	10	*Start any DC motor using corresponding starter and observe speed on tachometer.	2	CO3
LLO 11.1 Connect single phase induction motor to the supply. LLO 11.2 Change the direction of rotation of single phase induction.	11	*Start single phase induction motor and reverse the direction of rotation of it.	2	CO4
LLO 12.1 Connect the universal motor to the supply. LLO 12.2 Change the direction of rotation of universal motor.	12	Start universal motor and reverse the direction of rotation of it.	2	CO4
LLO 13.1 Connect the linear induction motor to the supply. LLO 13.2 Observe linear motion of induction motor.	13	Identify different parts of linear induction motor and start it.	2	CO4
LLO 14.1 Select fuse for particular application. LLO 14.2 Select circuit breaker for particular application.	14	*Identify different types of fuses and circuit breakers. State their specification for suitable application.	2	CO5
LLO 15.1 Explain connection of earthing for domestics application. LLO 15.2 Test available of earthing for given switch board.	15	Testing of earthing using a test lamp and comment on it.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- 1) Search the different types of applications in which a transformer is required and prepare a report on it.

2) Prepare a report on different types of applications of single-phase motors. State the types of motors with their particular applications.
3) Prepare a PowerPoint presentation or animation to show the working of the DC motor.
4) Observe the different types of switchgear used at home and write a report on their types, ratings and applications.

To build a simple electrical circuit

- 1) Construct a closed circuit using,

one dry cell battery, one small light bulb holder, one small light bulb, small wire
stripper tape (scotch, masking, or electrical)
Answer the following questions:
a) What is the difference between an open and a closed circuit?
b) What is voltage?
c) How many connections to the battery are necessary for the light bulb to light up?
2) Prepare a switchboard to control one lamp, one socket with protection and indication.

Micro project

- **1) Magnetic circuits**: Collect the information for different types of magnetic materials and draw a B-H curve for the respective material.

2) A.C. Fundamentals: Visit a nearby industry and observe the different parameters such as frequency, voltage, current, power and prepare a report based on it.
3) Polyphase circuits: Observe the three-phase power distribution panel in their institute and prepare a report on it.
4) Transformer: Collect information regarding different types of transformers available in the laboratory and prepare a report on it.
5) Fractional horsepower motor: Visit the local market or use the internet and prepare a report based on i) Manufacturers ii) Technical specifications iii) Earthing arrangement iv)Price range.
6) Visit your institute workshop and prepare a report on the different types of machines used, their specifications and manufacturers, different types of motors used.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Single phase 230V, 50Hz, 1Hp Induction motor	11
2	Single phase 230V, 50Hz, 1/4Hp Universal motor	12
3	Single or three phase linear induction motor	13
4	Single Phase 230/115 V, 50Hz, 1 or 5 kVA Transformer	2,7
5	Single Phase 0-270V, 50Hz, 10A Auto-transformer	2,7,11
6	Cathode Ray Oscilloscope (CRO) 20MHz, Dual channel	3,4,8
7	Single phase 230V, 50Hz, 2A Inductive Load bank	4
8	Single phase 230V, 50Hz, 2A Capacitive Load bank	4
9	Single phase 230V, 10A Resistive Load bank	4,5,6

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
10	Pulse transformer 1:1:1 4503 or 1:1 4502	8
11	Different types of DC motor	9,10

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Magnetic circuits	CO1	8	4	4	4	12
2	II	A.C fundamentals for single phase and polyphase circuits	CO2	11	4	6	8	18
3	III	Transformers and DC motors	CO3	8	6	4	4	14
4	IV	Fractional horse power motors	CO4	10	4	4	6	14
5	V	Electrical protective devices	CO5	8	4	4	4	12
Grand Total				**45**	**22**	**22**	**26**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Each practical will be assessed considering - - 60% weightage to process and - 40% weightage to product Continuous assessment based on process and product related performance indicators, laboratory experience.

Summative Assessment (Assessment of Learning)

- End of semester exam based on observations and recording of the particular experiments

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	3	3	1	2	-	2			
CO2	2	3	2	-	2	3	2			
CO3	3	2	3	2	2	-	2			
CO4	2	2	3	3	2	2	2			
CO5	3	3	2	2	3	2	3			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Theraja B.L.	Electrical Technology Vol-I	S.Chand and Co., new Delhi, ISBN:9788121924405
2	Theraja B.L.	Electrical Technology Vol-II	S.Chand and Co., new Delhi, ISBN:9788121924375

Sr.No	Author	Title	Publisher with ISBN Number
3	V. N. Mittle and Arvind Mittal	Basic Electrical Engineering	McGraw Hill, New Delhi, ISBN:978-0070593572
4	U.A.Bakshi	Basic Electrical Engineering	Technical Publications, ISBN:9789333220392
5	DP Kothari and I J Nagrath	Basic Electrical Engineering	Mc Graw Hill, New Delhi, ISBN: 978-9353165727
6	J.B. Gupta	A Course in Electrical Installation Estimating & Costing	S.K. Kataria & Sons, ISBN: 978-93-5014-279-0
7	K. B. Raina and S. K. Bhattacharya	Electrical design, estimation and costing, Second edition	New age international limited publisher, New Delhi, ISBN:978-8122443585

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://nptel.ac.in/courses/108105112	NPTEL study materials
2	https://www.electrical4U.com	All about electrical circuits
3	https://instrumentationtools.com/category/electrical-animation/	Animation of basic electrical engineering quantities
4	https://www.udemy.com/course/crash-course-electric-circuits-for-electrical-engineering/	Flip classroom learning material
5	http://www.ece.umn.edu/users/riaz/animations/listanimations.html	Animation of electrical machines
6	https://www.services.bis.gov.in/php/BIS_2.0/bisconnect/get_is_list_by_category_id/5	IS standard

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students